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CUET UG

Mathematics

Relations and Functions

Equivalence Relations, Mappings, Composition, Invertible Functions, Binary Operations & Inverse Trigonometry · Section B1 Unit I & II

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Aligned to the latest official syllabus · Concept-first explanations

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Syllabus Alignment & Exam Blueprint

Sub-Topic / Domain	Official NTA CUET UG Syllabus Scope (Mathematics 319)	Exam Cognitive Focus
Types of Relations	Binary relations on a set; Empty relation, Universal relation, Identity relation; Reflexive, Symmetric, and Transitive relations; Equivalence relations, Equivalence classes, and Partitions of sets.	Verifying the 3 equivalence axioms; identifying missing self-loops; computing equivalence classes; combinatorial counting formulas ($2^{\{n^2\}}, 2^{\{n(n-1)\}}$).
Function Mappings	One-to-one (Injective) and Many-one functions; Onto (Surjective) and Into functions; Bijective functions; Horizontal line test; Monotonicity calculus test; Counting injective/surjective functions.	Classifying mappings via algebra and derivatives; identifying domain/codomain mismatches; computing the number of bijections ($n!$) and surjections.
Composition & Inverses	Composition of functions $f \circ g$; Domain restrictions for composition; Associativity and non-commutativity; Invertible functions; Necessary and sufficient condition for invertibility (f is bijective); Reversal law $(f \circ g)^{-1} = g^{-1} \circ f^{-1}$.	Evaluating composite functions inside-out; determining algebraic formulas for $f^{-1}(x)$; geometric reflection across $y = x$.
Binary Operations	Definition $* : A \times A \to A$; Commutative and Associative properties; Identity element e; Inverse element a^{-1}; Cayley operation tables; Counting commutative operations.	Solving $a * e = a$ for identity; computing inverses; evaluating non-standard algebraic operations on $\mathbb{Q}$ and $\mathbb{R}$.
Inverse Trigonometric Functions	Definition, Domain, Range, and Principal Value Branches (PVB) of $\sin^{-1}, \cos^{-1}, \tan^{-1}, \cot^{-1}, \sec^{-1}, \csc^{-1}$; Graphs and monotonicity; Negative angle rules (Odd vs π - rule); Co-function identities ($\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$); Addition theorems for $\tan^{-1}$; Multiple angle transformations ($2\tan^{-1} x$).	Evaluating angles outside principal branches (sawtooth reductions); evaluating nested trig-inverse expressions; domain of definition problems; algebraic substitutions.

EXAM PATTERN

CUET UG Scoring Blueprint: Relations, Functions, and Inverse Trigonometry constitute the core foundational language of the CUET UG Mathematics examination (Paper 319), contributing **4 to 6 questions (20 to 30 marks)** across Section A and Section B1. NTA questions test **exact branch definitions** ($[0, \pi]$ vs $[-\frac{\pi}{2}, \frac{\pi}{2}]$), **unclosed transitivity chains**, **range-codomain matching for surjectivity**, and **boundary conditions for $\tan^{-1}$ sum identities** ($xy < 1$).

Relations & Equivalence Architecture

1. Cartesian Product & Binary Relations

Let A and B be two non-empty sets. The **Cartesian Product** $A \times B$ is the set of all ordered pairs: $A \times B = \{(a, b) : a \in A \text{ and } b \in B\}$. A **binary relation** R from A to B is any arbitrary subset of the Cartesian product: $R \subseteq A \times B$. If $|A| = m$ and $|B| = n$, then $|A \times B| = mn$, yielding exactly 2^{mn} total relations. On a single set A with n elements, there are 2^{n^2} total relations.

2. Fundamental Types of Relations

- **Empty Relation (φ):** $R = \varphi \subseteq A \times A$. Symmetric and transitive (vacuously true), but **never reflexive** on non-empty A.
- **Universal Relation ($A \times A$):** $R = A \times A$. Both empty and universal relations are **trivial relations**.
- **Identity Relation (I_A or Δ):** Every element relates strictly and exclusively to itself: $I_A = \{(a, a) : a \in A\}$. Always an equivalence relation.

3. The Three Equivalence Axioms

To evaluate a relation R on set A, we test three independent logical properties in strict sequence:

1. **Reflexivity:** $(a, a) \in R$ for **every** element $a \in A$. If even one element x lacks (x, x), reflexivity fails completely.
2. **Symmetry:** Whenever $(a, b) \in R$, it must necessarily follow that $(b, a) \in R$. Every forward pair requires its mirror reverse.
3. **Transitivity:** Whenever $(a, b) \in R$ and $(b, c) \in R$, it must imply $(a, c) \in R$. If no two-step chain exists, transitivity is vacuously satisfied!

Relation Type	Mathematical Requirement	Canonical NCERT Flagship Example
Reflexive	$(a, a) \in R, \quad \forall a \in A$	Congruence of triangles ($T_1 \cong T_1$ on set of all triangles).
Symmetric	$(a, b) \in R \Rightarrow (b, a) \in R$	Perpendicularity of straight lines ($L_1 \perp L_2 \Rightarrow L_2 \perp L_1$).
Transitive	$(a, b) \in R$ and $(b, c) \in R \Rightarrow (a, c) \in R$	Parallelism of straight lines ($L_1 \parallel L_2$ and $L_2 \parallel L_3 \Rightarrow L_1 \parallel L_3$).
Equivalence	Reflexive + Symmetric + Transitive	Divisibility: $R = \{(a, b) : a - b$ is divisible by $n\}$ on $\mathbb{Z}$.

4. Equivalence Classes & Partitions of a Set

Let R be an equivalence relation on set A. The **Equivalence Class** containing a is: $[a] = \{x \in A : (x, a) \in R\}$. **Fundamental Theorem:** Any two equivalence classes are either **completely identical** or **mutually disjoint** ($[a] = [b]$ or $[a] \cap [b] = \varphi$). The union of all equivalence classes reconstitutes A ($\cup_a [a] = A$). Thus, R partitions A into disjoint subsets.

5. Combinatorial Counting of Relations on Set A with n Elements

- **Total Relations:** 2^{n^2} | **Reflexive Relations:** $2^{n(n-1)}$ (diagonal n pairs mandatory)
- **Symmetric Relations:** $2^{\frac{n(n+1)}{2}}$ | **Reflexive and Symmetric Relations:** $2^{\frac{n(n-1)}{2}}$

Functions, Mappings & Classification Engine

1. Function Definition & Notation

A relation f from set A to set B is a **function** (or mapping, written $f : A \to B$) if and only if:

1. **Totality:** Every element $x \in A$ is associated with some element $y \in B$.
2. **Uniqueness:** No element $x \in A$ is associated with more than one element in B. That is:
$$(x, y_1) \in f \text{ and } (x, y_2) \in f \Rightarrow y_1 = y_2$$

Set A is the **Domain**, set B is the **Codomain**, and the collection of all actual output values is the **Range**:
$$\text{Range}(f) = \{f(x) : x \in A\} \subseteq \text{Codomain}(B)$$

2. Taxonomy of Functions

Injective Functions (One-One / Injection)

A function $f : A \to B$ is **one-one** if distinct elements in the domain have distinct images in the codomain:
$$x_1 \neq x_2 \Rightarrow f(x_1) \neq f(x_2) \quad \equiv \quad f(x_1) = f(x_2) \Rightarrow x_1 = x_2$$

- **Geometric Test (Horizontal Line Test):** If every horizontal line $y = k$ intersects the graph of $f(x)$ at **most once**, the function is one-one. If any horizontal line cuts the curve at two or more points, it is **many-one**.
- **Calculus Monotonicity Test:** If $f(x)$ is continuous and differentiable on an interval I, and either:
$$f'(x) > 0 \quad \forall x \in I \quad \text{(strictly increasing)} \quad \text{or} \quad f'(x) < 0 \quad \forall x \in I \quad \text{(strictly decreasing)}$$
then $f(x)$ is strictly monotonic and therefore **guaranteed to be one-one**!

Surjective Functions (Onto / Surjection)

A function $f : A \to B$ is **onto** if every element of the codomain B has at least one pre-image in the domain A:
$$\forall y \in B, \quad \exists x \in A \quad \text{such that} \quad f(x) = y \quad \iff \quad \text{Range}(f) = \text{Codomain}(B)$$

- **Algorithm to Test Surjectivity:**
  1. Let $y \in B$. Set $y = f(x)$ and solve algebraically for x in terms of y: $x = g(y)$.
  2. Verify whether this expression for x is real and belongs to the stated domain A for **every** $y \in B$.
  3. If any element $y \in B$ yields an undefined or out-of-domain x, the function is **into** (not onto).

Bijective Functions (One-One & Onto / Bijection)

A function $f : A \to B$ that is **both injective and surjective** is termed a **bijective function**.

- **Theorem of Invertibility:** A function possesses an inverse function $f^{-1} : B \to A$ **if and only if it is a bijection**.
- **Finite Set Cardinality Rule:** For finite sets A and B where $|A| = |B| = n$:
$$f : A \to B \text{ is one-one} \iff f \text{ is onto} \iff f \text{ is bijective}$$

3. Combinatorial Counting of Functions ($|A| = m, |B| = n$)

- **Total Functions:** Each of the m inputs independently has n choices:
$$N_{\text{total}} = n^m$$
- **Injective Functions (One-One):**
  - If $m > n$: $N_{\text{inj}} = 0$ (by the Pigeonhole Principle, some elements must share images).
  - If $m \leq n$: Permutations of selecting m distinct images from n targets:
$$N_{\text{inj}} = {}^{n}P_{m} = \frac{n!}{(n-m)!}$$
- **Bijective Functions (Bijections):** Exists only when $m = n$:
$$N_{\text{bij}} = n!$$
- **Surjective Functions (Onto):**
  - If $m < n$: $N_{\text{surj}} = 0$ (cannot cover a larger codomain).
  - If $m \geq n$: Derived via Principle of Inclusion-Exclusion:
$$N_{\text{surj}} = \sum_{r=1}^{n} (-1)^{n-r} \binom{n}{r} r^m$$

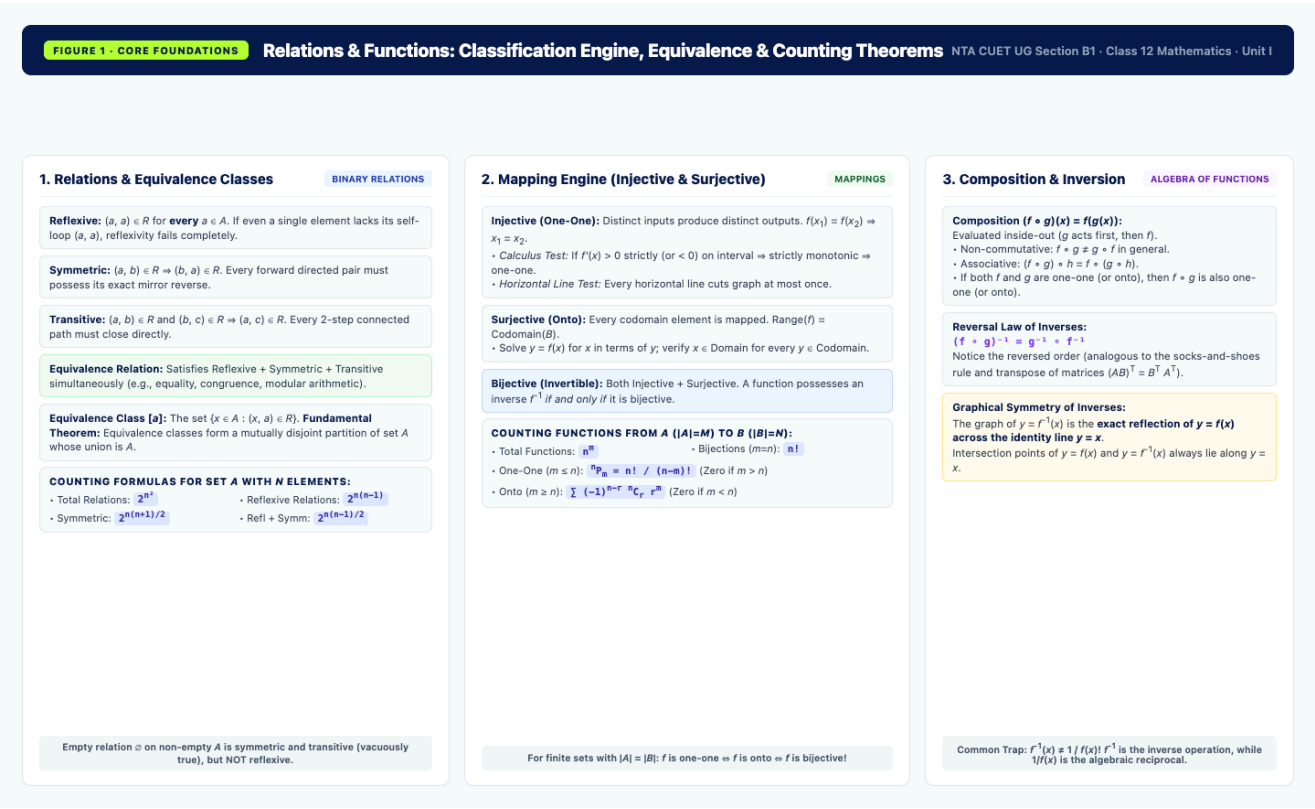


Figure 1: Figure 1: High-yield NCERT analytical framework — Relations classification (reflexive, symmetric, transitive, equivalence partitions), mapping taxonomy (injective, surjective, bijective), and combinatorial counting formulas.

Key Analytical Takeaways from Figure 1:

1. **Reflexivity Trap:** Reflexivity demands that **every single element** in set A has its diagonal pair $(a, a) \in R$. If set $A = \{1, 2, 3\}$, missing even $(3, 3)$ instantly invalidates reflexivity, regardless of how many other pairs exist.
2. **Monotonicity Principle:** For any differentiable function, verifying that the first derivative does not change sign ($f'(x) > 0$ or $f'(x) < 0$ strictly) proves injectivity without laborious algebraic manipulation.
3. **Bijection as Invertibility License:** A function cannot be inverted if it is many-one (the reverse mapping would fail uniqueness) or into (the reverse mapping would fail totality). Both conditions are mandatory.

Composition of Functions & Invertibility

1. Composition of Functions

Let $g : A \to B$ and $f : B \to C$ be two functions. The **composite function** $f \circ g : A \to C$ is defined by:

$$(f \circ g)(x) = f(g(x)) \quad \forall x \in A$$

- **Existence Criterion:** The composition $f \circ g$ is well-defined if and only if the range of the inner function g is a subset of the domain of the outer function f:

$$\text{Range}(g) \subseteq \text{Domain}(f)$$

- **Evaluation Order:** Composition is evaluated **inside-out**. The input x is first transformed by g, and the resulting output $g(x)$ is subsequently transformed by f.

Key Theorems on Composition

1. **Non-Commutativity:** In general, composition is **not commutative**:

$$f \circ g \neq g \circ f$$

Even when both compositions exist, their outputs, domains, and formulas usually differ completely.

2. **Associativity:** Function composition is always **associative**:

$$(f \circ g) \circ h = f \circ (g \circ h)$$

3. **Preservation of Invariance:**
   - If both f and g are one-one, then $f \circ g$ is **one-one**.
   - If both f and g are onto, then $f \circ g$ is **onto**.
   - If both f and g are bijections, then $f \circ g$ is a **bijection**.
4. **Deductive Implications (High-Yield NTA Trap):**
   - If $g \circ f$ is **one-one**, then the inner function f must be **one-one** (outer function g may or may not be one-one).
   - If $g \circ f$ is **onto**, then the outer function g must be **onto** (inner function f may or may not be onto).

2. Invertible Functions

A function $f : X \to Y$ is defined to be **invertible** if there exists a function $g : Y \to X$ such that:

$$g \circ f = I_X \quad \text{and} \quad f \circ g = I_Y$$

Where I_X and I_Y are the identity functions on sets X and Y respectively. The function g is uniquely determined and is termed the **inverse of f**, denoted by f^{-1}.

$$f(x) = y \iff f^{-1}(y) = x$$

Reversal Law of Inverses

Let $f : X \to Y$ and $g : Y \to Z$ be two bijective functions. Then the composite function $g \circ f : X \to Z$ is invertible, and its inverse obeys the fundamental **Reversal Law**:

$$(g \circ f)^{-1} = f^{-1} \circ g^{-1}$$

Intuitive Logic (The Socks-and-Shoes Rule): If you put on socks (f) first and then put on shoes (g), to undo the process, you must remove shoes (g^{-1}) first and then remove socks (f^{-1}).

Geometric Reflection Principle

If (a, b) lies on the curve $y = f(x)$, then by definition $f(a) = b$, which implies $f^{-1}(b) = a$. Thus, the point (b, a) lies on the curve $y = f^{-1}(x)$.

- The points (a, b) and (b, a) are mutual mirror reflections across the line $y = x$.
- Consequently, the **graph of $y = f^{-1}(x)$ is the exact reflection of the graph of $y = f(x)$ about the identity diagonal $y = x$**.
- Any points where a function intersects its own inverse must lie along the line of symmetry $y = x$ (i.e., $f(x) = x$).

Inverse Trigonometric Functions: The Six Branches

1. Why Restrict Domains?

The classical trigonometric functions ($\sin x, \cos x, \tan x, ...$) are periodic functions over $\mathbb{R}$, meaning they repeat their values every 2π (or π) radians. Because multiple inputs produce identical outputs (many-one), trigonometric functions are **not invertible over their natural domains**.

- To construct inverse functions, we strictly restrict their domains to intervals where they are **strictly monotonic and bijective**.
- The standard chosen intervals are internationally codified as the **Principal Value Branches (PVB)**.

2. Systematic Audit of the Six Principal Branches

Inverse Function	Domain (Inputs x)	Principal Value Range (Outputs y)	Calculus & Graph Nature
$y = \sin^{-1} x$	$[-1, 1]$	$[-\frac{\pi}{2}, \frac{\pi}{2}]$	Strictly increasing; passes $(0, 0)$; odd symmetry.
$y = \cos^{-1} x$	$[-1, 1]$	$[0, \pi]$	Strictly decreasing; always $y \geq 0$; no origin crossing.
$y = \tan^{-1} x$	$(-\infty, \infty) = \mathbb{R}$	$(-\frac{\pi}{2}, \frac{\pi}{2})$ **(open)**	Strictly increasing; asymptotes $y = \pm\frac{\pi}{2}$.
$y = \cot^{-1} x$	$(-\infty, \infty) = \mathbb{R}$	$(0, \pi)$ **(open)**	Strictly decreasing; strictly positive range.
$y = \sec^{-1} x$	$(-\infty, -1] \cup [1, \infty)$	$[0, \pi] - \{\frac{\pi}{2}\}$	Discontinuous at $\lvert x\rvert < 1$; undefined at $y = \frac{\pi}{2}$.
$y = \csc^{-1} x$	$(-\infty, -1] \cup [1, \infty)$	$[-\frac{\pi}{2}, \frac{\pi}{2}] - \{0\}$	Discontinuous at $\lvert x\rvert < 1$; undefined at $y = 0$.

3. Self-Composition & The Sawtooth Phenomenon

Examiners frequently test expressions of the form $f^{-1}(f(x))$ and $f(f^{-1}(x))$:

1. **Outer Trigonometric Function:**

$$\sin(\sin^{-1} x) = x \quad \text{for all } x \in [-1, 1]$$
$$\cos(\cos^{-1} x) = x \quad \text{for all } x \in [-1, 1]$$
$$\tan(\tan^{-1} x) = x \quad \text{for all } x \in \mathbb{R}$$

   This identity holds universally across the entire domain of the inverse function without exception.

2. **Outer Inverse Function (The Principal Branch Trap):**

$$\sin^{-1}(\sin x) = x \quad \text{ONLY IF } x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$$
$$\cos^{-1}(\cos x) = x \quad \text{ONLY IF } x \in [0, \pi]$$
$$\tan^{-1}(\tan x) = x \quad \text{ONLY IF } x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$$

   If the angle x lies outside the principal branch, you **must not cancel $\sin^{-1}$ with sin**! Instead, use periodicity and quadrant angle shifts:

   - For $\sin^{-1}(\sin x)$:
     - In Quadrant II ($\frac{\pi}{2} \leq x \leq \frac{3\pi}{2}$): $\sin^{-1}(\sin x) = \pi - x$
     - In Quadrant III/IV ($\frac{3\pi}{2} \leq x \leq \frac{5\pi}{2}$): $\sin^{-1}(\sin x) = x - 2\pi$
   - For $\cos^{-1}(\cos x)$:
     - For $\pi \leq x \leq 2\pi$: $\cos^{-1}(\cos x) = 2\pi - x$

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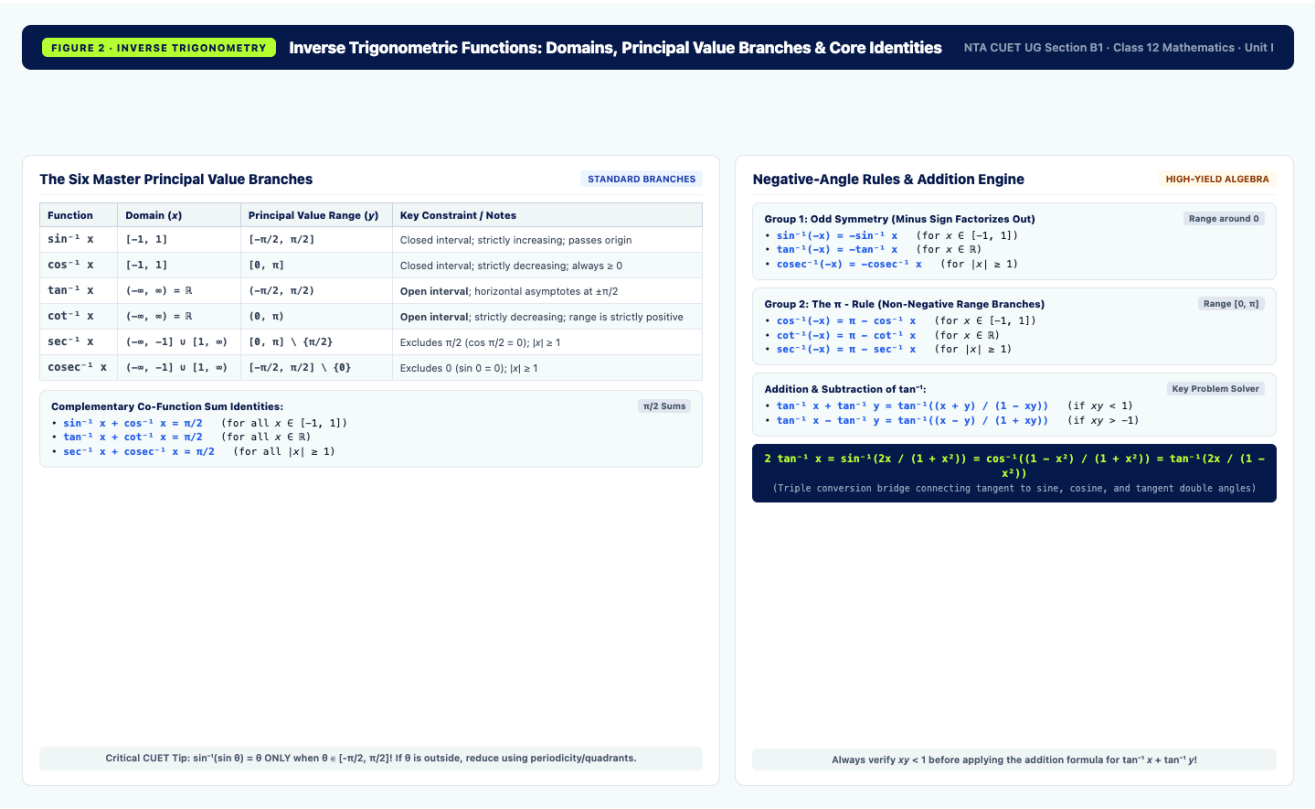


Figure 2: Figure 2: Systematic comparison matrix — Inverse Trigonometric Functions: Domains, Principal Value Branches, Odd vs π-Rule Negative Angle Identities, and the $\tan^{-1}$ Addition Engine.

Key Analytical Takeaways from Figure 2:

1. **Open vs Closed Intervals:** $\tan^{-1} x$ maps to the **open** interval $(-\frac{\pi}{2}, \frac{\pi}{2})$ because $\tan(\pm\frac{\pi}{2}) = \pm\infty$ is undefined. Selecting a closed bracket option $[-\frac{\pi}{2}, \frac{\pi}{2}]$ is one of the most common NTA traps.
2. **Two Negative-Angle Regimes:**
   - **Odd Symmetry Group ($\sin^{-1}, \tan^{-1}, \csc^{-1}$):** Negative sign pulls directly outside: $\sin^{-1}(-x) = -\sin^{-1} x$.
   - **π - Subtraction Group ($\cos^{-1}, \cot^{-1}, \sec^{-1}$):** Because ranges are non-negative ($[0, \pi]$ or $(0, \pi)$), negative inputs reflect across $\frac{\pi}{2}$: $\cos^{-1}(-x) = \pi - \cos^{-1} x$.
3. **Co-Function Sum Identities:** The three complementary sums ($\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$, $\tan^{-1} x + \cot^{-1} x = \frac{\pi}{2}$, $\sec^{-1} x + \csc^{-1} x = \frac{\pi}{2}$) hold identically across their valid domains and instantly simplify complex algebraic sums.

Advanced Identities & Algebraic Toolkit of ITF

1. Negative-Angle Transformation Identities

The six inverse trigonometric functions split strictly into two symmetric families:

1. Negative-Angle Transformation Identities

The six inverse trigonometric functions split strictly into two symmetric families:

- **Odd Symmetry Family (Minus Pulls Out Directly):** Range centered around 0.
$$\sin^{-1}(-x) = -\sin^{-1} x, \quad \tan^{-1}(-x) = -\tan^{-1} x, \quad \csc^{-1}(-x) = -\csc^{-1} x$$
- **π - Shift Rule Family (Non-Negative Range Branches):** Range spans $[0, \pi]$ or $(0, \pi)$.
$$\cos^{-1}(-x) = \pi - \cos^{-1} x, \quad \cot^{-1}(-x) = \pi - \cot^{-1} x, \quad \sec^{-1}(-x) = \pi - \sec^{-1} x$$

2. Complementary Co-Function Sum Identities

When arguments are identical and lie within the valid domain:

$$\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2} \quad (x \in [-1, 1]); \quad \tan^{-1} x + \cot^{-1} x = \frac{\pi}{2} \quad (x \in \mathbb{R}); \quad \sec^{-1} x + \csc^{-1} x = \frac{\pi}{2} \quad (|x| \geq 1)$$

3. Reciprocal Identities

$$\sin^{-1}\left(\frac{1}{x}\right) = \csc^{-1} x \quad (|x| \geq 1); \quad \cos^{-1}\left(\frac{1}{x}\right) = \sec^{-1} x \quad (|x| \geq 1); \quad \tan^{-1}\left(\frac{1}{x}\right) = \begin{cases} \cot^{-1} x & \text{if } x > 0 \\ -\pi + \cot^{-1} x & \text{if } x < 0 \end{cases}$$

4. Addition & Subtraction Theorems for $\tan^{-1}$

- **Sum Formula:** $\tan^{-1} x + \tan^{-1} y = \begin{cases} \tan^{-1}\left(\frac{x+y}{1-xy}\right) & \text{if } xy<1 \\ \pi + \tan^{-1}\left(\frac{x+y}{1-xy}\right) & \text{if } xy>1 \text{ and } x>0 \\ -\pi + \tan^{-1}\left(\frac{x+y}{1-xy}\right) & \text{if } xy>1 \text{ and } x<0 \end{cases}$
- **Difference Formula:** $\tan^{-1} x - \tan^{-1} y = \tan^{-1}\left(\frac{x-y}{1+xy}\right)$ for $xy > -1$.

5. The Multiple-Angle Triple Conversion Bridge

The expression $2\tan^{-1} x$ connects tangent double-angle forms:

$$2\tan^{-1} x = \sin^{-1}\left(\frac{2x}{1+x^2}\right) \quad (|x| \leq 1) = \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right) \quad (x \geq 0) = \tan^{-1}\left(\frac{2x}{1-x^2}\right) \quad (-1 < x < 1)$$

6. High-Yield Substitution Toolkit for Algebraic Expressions

- For $\sqrt{a^2 - x^2}$: $x = a\sin\theta$ or $a\cos\theta$ | For $\sqrt{a^2 + x^2}$: $x = a\tan\theta$ or $a\cot\theta$
- For $\sqrt{x^2 - a^2}$: $x = a\sec\theta$ or $a\csc\theta$ | For $\sqrt{\frac{a-x}{a+x}}$: $x = a\cos 2\theta$

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Binary Operations (CUET Applied & Core Mathematics)

1. Mathematical Definition of a Binary Operation

A **binary operation** $*$ on a non-empty set A is a function that maps every ordered pair of elements in $A \times A$ to an element in A:

$$* : A \times A \to A$$

We write $*(a, b)$ as $a * b$.

- **Closure Property:** For $*$ to be a binary operation on A, it is mandatory that for all $a, b \in A$, the result $a * b$ must belong to A (the set is closed under $*$).
- **Example:** Addition and multiplication are binary operations on $\mathbb{N}$, but subtraction and division are not (e.g., $3 - 5 = -2 \notin \mathbb{N}$).

2. Core Algebraic Properties

1. **Commutative Property:** A binary operation $*$ on A is **commutative** if:
$$a * b = b * a \quad \forall a, b \in A$$
2. **Associative Property:** A binary operation $*$ on A is **associative** if:
$$(a * b) * c = a * (b * c) \quad \forall a, b, c \in A$$
Associativity permits writing expressions like $a * b * c$ unambiguously without brackets.

3. Identity & Inverse Elements

Identity Element (e)

An element $e \in A$ is called the **identity element** for the binary operation $*$ if:

$$a * e = e * a = a \quad \forall a \in A$$

- **Uniqueness Theorem:** If an identity element exists for a binary operation, it is strictly **unique**.
- **Examples:** For addition on $\mathbb{R}$, $e = 0$; for multiplication on $\mathbb{R}$, $e = 1$.

Inverse Element (a^{-1})

Given a binary operation $*$ on A with identity element e, an element $a \in A$ is said to be **invertible** if there exists an element $b \in A$ such that:

$$a * b = b * a = e$$

Here, b is called the **inverse of a** and is denoted by a^{-1}.

- If b is the inverse of a, then a is the inverse of b: $(a^{-1})^{-1} = a$.
- **Reversal Law:** If $*$ is associative, $(a * b)^{-1} = b^{-1} * a^{-1}$.

4. Cayley Operation Tables on Finite Sets

For a finite set $A = \{a_1, a_2, ..., a_n\}$, a binary operation can be completely tabulated in an $n \times n$ matrix where the entry in row i and column j is $a_i * a_j$.

- **Symmetry Test:** The operation $*$ is commutative if and only if the Cayley table is completely **symmetric across its main diagonal**.
- **Identity Test:** An element e is the identity if its row and column match the header row and column identically.
- **Counting Binary Operations:**
  - Total binary operations on a set with n elements: n^{n^2}
  - Commutative binary operations: $n^{\frac{n(n+1)}{2}}$

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Key Facts & Terms Cheatsheet

MASTER DIAGNOSTIC & CONCEPTUAL CHEATSHEET — RELATIONS, FUNCTIONS & INVERSE TRIGONOMETRY

Key Mathematical Topic	Core NCERT Mechanism / Definition	Exam Identifier
Reflexive Relation	$(a, a) \in R$ for every single element $a \in A$. Missing one pair destroys it.	Every $(a, a) \in R$
Symmetric Relation	$(a, b) \in R \Rightarrow (b, a) \in R$. Every forward pair has its mirror reverse.	Two-Way Symmetry
Transitive Relation	$(a, b) \in R$ and $(b, c) \in R \Rightarrow (a, c) \in R$. Chains must close.	Chain Closes
Equivalence Relation	Reflexive + Symmetric + Transitive simultaneously (e.g. congruence, mod n).	All 3 Axioms Pass
Equivalence Classes	Mutually disjoint subsets $[a] = \{x : (x, a) \in R\}$ partitioning set A.	Disjoint Partition
Total Relations (n)	Number of subsets of $A \times A$: 2^{n^2}.	2^{n^2} Relations
Reflexive Relations	$2^{n(n-1)}$ relations (diagonal n pairs mandatory).	$2^{n(n-1)}$
Symmetric Relations	$2^{\frac{n(n+1)}{2}}$ relations (diagonal and upper triangle independent).	$2^{\frac{n(n+1)}{2}}$
Injective (One-One)	$f(x_1) = f(x_2) \Rightarrow x_1 = x_2$; horizontal line cuts graph at most once.	Distinct Outputs
Surjective (Onto)	$\text{Range}(f) = \text{Codomain}(B)$; every y has at least one pre-image x.	Range = Codomain
Bijective (Bijection)	Both one-one and onto. Unique condition for existence of inverse f^{-1}.	Invertible Mapping
Total Functions ($m \to n$)	n^m total mappings from domain $\lvert A\rvert = m$ to codomain $\lvert B\rvert = n$.	n^m Functions
One-One Count ($m \leq n$)	${}^{\{\}}n P_m = \frac{n!}{(n-m)!}$ injections. If $m > n$, count is zero.	${}^{\{\}}n P_m$
Bijection Count ($m = n$)	$n!$ bijections between finite sets of equal cardinality.	$n!$ Bijections
Reversal Law (Inverses)	$(f \circ g)^{-1} = g^{-1} \circ f^{-1}$ (undoing composite operations).	Reverse Order
Geometric Inverse	Graph of $y = f^{-1}(x)$ is mirror reflection of $y = f(x)$ across $y = x$.	Mirror Across $y = x$
$\sin^{-1} x$ Branch	Domain $[-1, 1]$, Principal Range $[-\frac{\pi}{2}, \frac{\pi}{2}]$, strictly increasing.	$[-\frac{\pi}{2}, \frac{\pi}{2}]$
$\cos^{-1} x$ Branch	Domain $[-1, 1]$, Principal Range $[0, \pi]$, strictly decreasing.	$[0, \pi]$
$\tan^{-1} x$ Branch	Domain $\mathbb{R}$, Principal Range $(-\frac{\pi}{2}, \frac{\pi}{2})$ (open brackets!), increasing.	Open Interval
π - Rule (ITF)	$\cos^{-1}(-x) = \pi - \cos^{-1} x$, $\cot^{-1}(-x) = \pi - \cot^{-1} x$, $\sec^{-1}(-x) = \pi - \sec^{-1} x$.	π Minus Group
Odd Symmetry (ITF)	$\sin^{-1}(-x) = -\sin^{-1} x$, $\tan^{-1}(-x) = -\tan^{-1} x$, $\csc^{-1}(-x) = -\csc^{-1} x$.	Minus Factorizes
Complementary Sum	$\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$, $\tan^{-1} x + \cot^{-1} x = \frac{\pi}{2}$, $\sec^{-1} x + \csc^{-1} x = \frac{\pi}{2}$.	Sum = $\frac{\pi}{2}$

Common Errors and High-Yield CUET Traps

COMMON STUDENT MISTAKES & DIAGNOSTIC TRAPS — RELATIONS, FUNCTIONS & ITF

Common Student Mistake	Scientifically Verified Reality	The Exact Trap Mechanism
Confusing Inverse with Reciprocal	$f^{-1}(x)$ denotes the **inverse function** (undoing the mapping), whereas $\frac{1}{f(x)} = (f(x))^{-1}$ is the arithmetic reciprocal.	Students select $\frac{1}{3x-4}$ when asked for the inverse of $f(x) = 3x - 4$. The inverse is $\frac{x+4}{3}$.
Checking Reflexivity on Listed Pairs Only	Reflexivity requires $(a, a) \in R$ for **every element in universal set A**, not just elements that appear in relation pairs.	On $A = \{1, 2, 3\}$, $R = \{(1, 1), (2, 2)\}$ is NOT reflexive because $(3, 3)$ is missing.
Blind Cancellation of $\sin^{-1}(\sin x)$	$\sin^{-1}(\sin x) = x$ holds **ONLY when $x \in [-\frac{\pi}{2}, \frac{\pi}{2}]$**. If $x = \frac{3\pi}{5}$, $\sin^{-1}(\sin(\frac{3\pi}{5})) = 2\frac{\pi}{5}$, NOT $\frac{3\pi}{5}$.	NTA offers $\frac{3\pi}{5}$ as Option A to trap students who cancel inverse with trig blindly.
Assuming $\cos^{-1}(-x) = -\cos^{-1} x$	The range of $\cos^{-1} x$ is $[0, \pi]$ (strictly non-negative). Thus $\cos^{-1}(-x) = \pi - \cos^{-1} x$, NOT $-\cos^{-1} x$.	Students confuse the odd symmetry of $\sin^{-1}$ with $\cos^{-1}$, selecting negative angles for $\cos^{-1}$.
Square Brackets on $\tan^{-1}$ Range	The range of $\tan^{-1} x$ is the **open interval** $(-\frac{\pi}{2}, \frac{\pi}{2})$. It can never attain $\pm\frac{\pi}{2}$ because $\tan(\pm\frac{\pi}{2})$ is undefined.	Option sets present both $[-\frac{\pi}{2}, \frac{\pi}{2}]$ and $(-\frac{\pi}{2}, \frac{\pi}{2})$; students miss the bracket difference.
Reversing Composition Order	$(f \circ g)(x) = f(g(x))$ means g operates **first**, and f operates **second**. It is generally unequal to $(g \circ f)(x)$.	NTA questions provide the value of $(g \circ f)(x)$ as an option for $(f \circ g)(x)$.
Applying $\tan^{-1}$ Sum When $xy \geq 1$	$\tan^{-1} x + \tan^{-1} y = \tan^{-1}\left(\frac{x+y}{1-xy}\right)$ requires **strictly $xy < 1$**. If $xy > 1$ and $x, y > 0$, you must add π.	Evaluating $\tan^{-1}(2) + \tan^{-1}(3)$ gives $\pi + \tan^{-1}(-1) = 3\frac{\pi}{4}$, NOT $-\frac{\pi}{4}$.
Violating Invertibility License	A function on $\mathbb{R}$ like $f(x) = x^2$ has no inverse because it is many-one ($f(-2) = f(2)$). Inverses require bijections.	Questions ask for f^{-1} of non-injective functions; students algebraically solve for x without checking domain injectivity.

OurMocks

The Whole Chapter on One Page

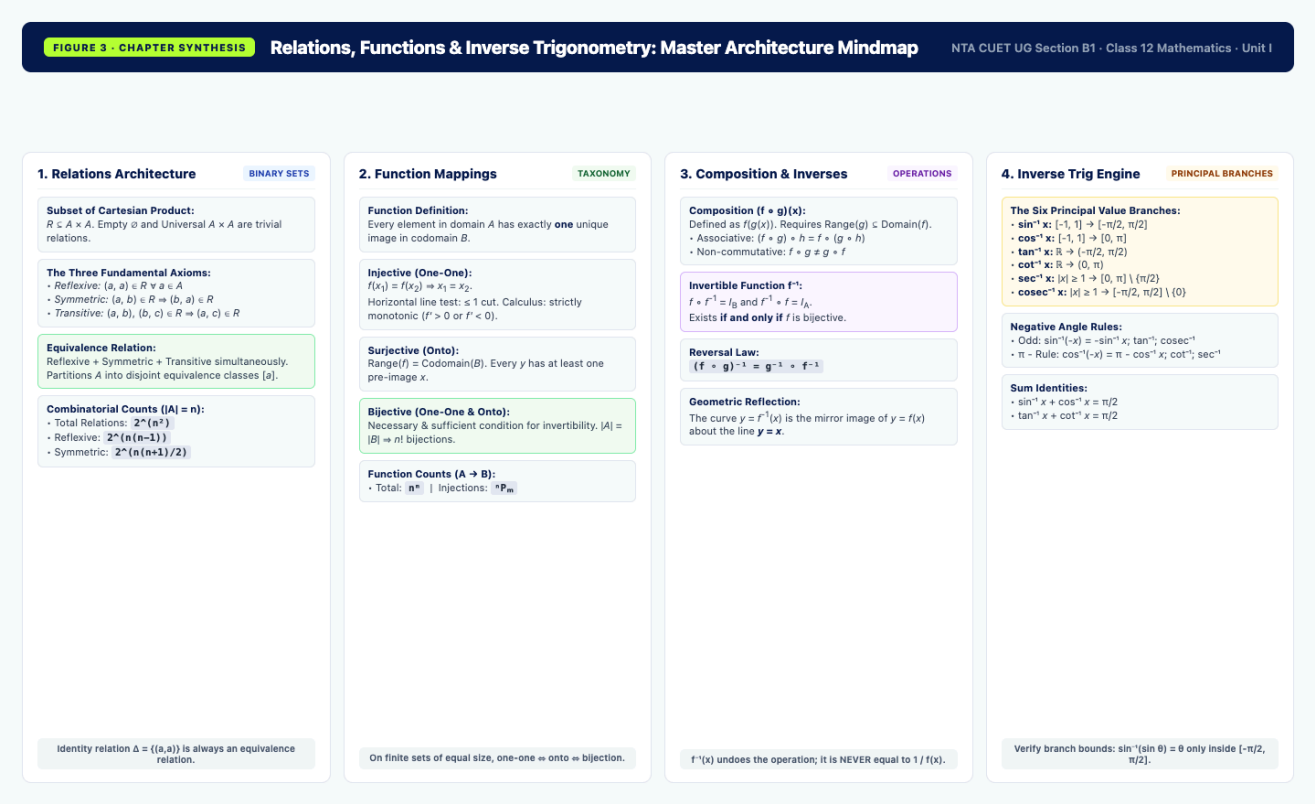


Figure 3: Figure 3: Master synthesis mindmap — high-yield integrated architecture of Relations, Functions, and Inverse Trigonometry covering relation axioms, mapping taxonomy, composition, and the six principal branches.

OurMocks

Practice Questions — Part 1 (Q1 to Q5)

Q1. Let $\mathbb{Z}$ be the set of integers and R be a relation on $\mathbb{Z}$ defined by $R = \{(a, b) : a - b \text{ is divisible by } 5\}$. Then R is:

(A) Reflexive and symmetric but not transitive
(B) Symmetric and transitive but not reflexive
(C) An equivalence relation with equivalence class $[0] = \{5k : k \in \mathbb{Z}\}$
(D) Transitive but neither reflexive nor symmetric

Sol. Q1. We test all three axioms for relation R:

1. **Reflexive:** For any $a \in \mathbb{Z}$, $a - a = 0 = 5 \times 0$, which is divisible by 5. Thus $(a, a) \in R$ for all $a \in \mathbb{Z}$.
2. **Symmetric:** Let $(a, b) \in R \Rightarrow a - b = 5k$ for some $k \in \mathbb{Z}$. Then $b - a = -(a - b) = 5(-k)$. Since $-k \in \mathbb{Z}$, $b - a$ is divisible by 5, so $(b, a) \in R$.
3. **Transitive:** Let $(a, b) \in R$ and $(b, c) \in R$. Then $a - b = 5k$ and $b - c = 5m$. Adding both yields $(a - b) + (b - c) = a - c = 5(k + m)$. Since $k + m \in \mathbb{Z}$, $a - c$ is divisible by 5, so $(a, c) \in R$.

Thus, R is an **equivalence relation**. The equivalence class of 0 is $[0] = \{x \in \mathbb{Z} : x - 0 = 5k\} = \{5k : k \in \mathbb{Z}\}$. **Answer: (C)**

Q2. If set A has 3 elements, then the number of reflexive relations, symmetric relations, and equivalence relations on A are respectively:

(A) 64, 64, 5
(B) 64, 32, 6
(C) 512, 64, 5
(D) 64, 512, 8

Sol. Q2. For a set A with $n = 3$ elements:

1. **Reflexive Relations:** The $n = 3$ diagonal pairs must be present (1 choice each). The remaining $n^2 - n = 9 - 3 = 6$ non-diagonal pairs each have 2 choices (present or absent). Total $= 2^{n(n-1)} = 2^6 = 64$.
2. **Symmetric Relations:** Choices are made independently for the 3 diagonal pairs and $\frac{n(n-1)}{2} = 3$ pairs above the diagonal. Total $= 2^{\frac{n(n+1)}{2}} = 2^{3 \times \frac{4}{2}} = 2^6 = 64$.
3. **Equivalence Relations:** The number of equivalence relations on a set with n elements equals the Bell number B_n. For $n = 3$, $B_3 = 5$ (corresponding to partitions: 3 elements together, $2 + 1$ in 3 ways, and 3 singletons).

Answer: (A)

Q3. Let $A = \mathbb{R} - \{3\}$ and $B = \mathbb{R} - \{1\}$. Consider the function $f : A \to B$ defined by $f(x) = \frac{x-2}{x-3}$. The function f is:

(A) One-one but not onto
(B) Onto but not one-one
(C) Neither one-one nor onto
(D) Bijective (both one-one and onto)

Sol. Q3.

1. **Injectivity (One-one):** Let $x_1, x_2 \in A$. Then:

$$\frac{x_1 - 2}{x_1 - 3} = \frac{x_2 - 2}{x_2 - 3} \Rightarrow (x_1 - 2)(x_2 - 3) = (x_2 - 2)(x_1 - 3)$$

$$x_1x_2 - 3x_1 - 2x_2 + 6 = x_1x_2 - 3x_2 - 2x_1 + 6 \Rightarrow -3x_1 - 2x_2 = -3x_2 - 2x_1 \Rightarrow -x_1 = -x_2 \Rightarrow x_1 = x_2$$

Hence, f is strictly **one-one**.

2. **Surjectivity (Onto):** Let $y \in B = \mathbb{R} - \{1\}$. We solve $y = \frac{x-2}{x-3}$ for x:

$$y(x - 3) = x - 2 \Rightarrow xy - 3y = x - 2 \Rightarrow x(y - 1) = 3y - 2 \Rightarrow x = \frac{3y - 2}{y - 1}$$

Since $y \neq 1$, x is well-defined and real. If $x = 3$, then $3(y - 1) = 3y - 2 \Rightarrow -3 = -2$ (impossible). Thus $x \in A$ for all $y \in B$. Range $= B$, so f is **onto**. Being both one-one and onto, f is **bijective**. **Answer: (D)**

Q4. Let set $A = \{1, 2, 3\}$ and set $B = \{p, q, r, s\}$. The total number of injective (one-one) functions that can be defined from A to B is:

(A) 64
(B) 24
(C) 12
(D) 81

Sol. Q4. Here $|A| = m = 3$ and $|B| = n = 4$. An injective function requires that each element in A maps to a distinct element in B.

- The first element of A has 4 choices in B.
- The second element of A has 3 remaining choices in B.
- The third element of A has 2 remaining choices in B.

By the fundamental counting principle:

$$N_{\text{inj}} = \{\}^nP_m = \{\}^4P_3 = \frac{4!}{(4-3)!} = \frac{24}{1} = 24$$

(Note: Total functions $= n^m = 4^3 = 64$.) **Answer: (B)**

Q5. Let $f(x) = \sqrt{x}$ and $g(x) = 2x - 3$. The domain of the composite function $(f \circ g)(x)$ is:

(A) $\mathbb{R}$
(B) $[0, \infty)$
(C) $\left[\frac{3}{2}, \infty\right)$
(D) $\left(-\infty, \frac{3}{2}\right]$

Sol. Q5. By definition of composite function:

$$(f \circ g)(x) = f(g(x)) = f(2x - 3) = \sqrt{2x - 3}$$

For the square root function to yield real values, the radicand must be non-negative:

$$2x - 3 \geq 0 \Rightarrow 2x \geq 3 \Rightarrow x \geq \frac{3}{2}$$

Thus, the domain of $(f \circ g)$ is $\left[\frac{3}{2}, \infty\right)$. **Answer: (C)**

Practice Questions — Part 2 (Q6 to Q10)

Q6. Let $f : [1, \infty) \to [2, \infty)$ be given by $f(x) = x^2 - 2x + 3$. The inverse function $f^{-1}(x)$ is:

(A) $1 + \sqrt{x-2}$ **(B)** $1 - \sqrt{x-2}$
(C) $\sqrt{x-3}+1$ **(D)** Does not exist because f is quadratic

Sol. Q6. Complete the square on $f(x)$:

$$y = x^2 - 2x + 3 = (x-1)^2 + 2$$

$$(x-1)^2 = y - 2 \Rightarrow x - 1 = \pm\sqrt{y-2}$$

Since the domain is restricted to $x \in [1, \infty)$, we have $x - 1 \geq 0$, meaning we must select the positive branch:

$$x = 1 + \sqrt{y-2}$$

Swapping variables gives the inverse function:

$$f^{-1}(x) = 1 + \sqrt{x-2}$$

Note: On the unrestricted domain $\mathbb{R}$, f is many-one; but on $[1, \infty)$, f is strictly increasing ($f'(x) = 2x - 2 \geq 0$) and surjective onto $[2, \infty)$, making it fully invertible. **Answer: (A)**

Q7. The principal value of $\cos^{-1}\left(\cos\left(\frac{7\pi}{6}\right)\right)$ is equal to:

(A) $\frac{7\pi}{6}$ **(B)** $\frac{5\pi}{6}$
(C) $\frac{\pi}{6}$ **(D)** $-\frac{\pi}{6}$

Sol. Q7. The principal value branch of $\cos^{-1} x$ is $[0, \pi]$. Since $\frac{7\pi}{6} = 210° \notin [0, \pi]$, we cannot blindly cancel $\cos^{-1}$ with cos. Using the cosine periodicity and quadrant reduction:

$$\cos\left(\frac{7\pi}{6}\right) = \cos\left(2\pi - \frac{5\pi}{6}\right) = \cos\left(\frac{5\pi}{6}\right)$$

Now, $\frac{5\pi}{6} = 150° \in [0, \pi]$. Therefore:

$$\cos^{-1}\left(\cos\left(\frac{7\pi}{6}\right)\right) = \cos^{-1}\left(\cos\left(\frac{5\pi}{6}\right)\right) = \frac{5\pi}{6}$$

Answer: (B)

Q8. The principal value of $\sin^{-1}\left(\sin\left(\frac{3\pi}{5}\right)\right)$ is equal to:

(A) $\frac{3\pi}{5}$ **(B)** $-\frac{3\pi}{5}$
(C) $\frac{2\pi}{5}$ **(D)** $\frac{\pi}{5}$

Sol. Q8. The principal value branch of $\sin^{-1} x$ is $[-\frac{\pi}{2}, \frac{\pi}{2}]$. Here $\frac{3\pi}{5} = 108° \notin [-90°, 90°]$. Using the supplementary angle identity for sine ($\sin(\pi - \theta) = \sin\theta$):

$$\sin\left(\frac{3\pi}{5}\right) = \sin\left(\pi - \frac{2\pi}{5}\right) = \sin\left(\frac{2\pi}{5}\right)$$

Now check the angle: $\frac{2\pi}{5} = 72° \in [-90°, 90°] = [-\frac{\pi}{2}, \frac{\pi}{2}]$. Therefore:

$$\sin^{-1}\left(\sin\left(\frac{3\pi}{5}\right)\right) = \sin^{-1}\left(\sin\left(\frac{2\pi}{5}\right)\right) = \frac{2\pi}{5}$$

Answer: (C)

Q9. If $\sin^{-1} x + \sin^{-1} y = \frac{2\pi}{3}$, then the value of $\cos^{-1} x + \cos^{-1} y$ is:

(A) $\frac{2\pi}{3}$ **(B)** $\frac{\pi}{3}$
(C) $\frac{\pi}{6}$ **(D)** π

Sol. Q9. Recall the complementary co-function sum identity:

$$\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2} \Rightarrow \cos^{-1} x = \frac{\pi}{2} - \sin^{-1} x$$

$$\sin^{-1} y + \cos^{-1} y = \frac{\pi}{2} \Rightarrow \cos^{-1} y = \frac{\pi}{2} - \sin^{-1} y$$

Adding both equations:

$$\cos^{-1} x + \cos^{-1} y = \left(\frac{\pi}{2} + \frac{\pi}{2}\right) - (\sin^{-1} x + \sin^{-1} y) = \pi - (\sin^{-1} x + \sin^{-1} y)$$

Substitute the given value $\sin^{-1} x + \sin^{-1} y = \frac{2\pi}{3}$:

$$\cos^{-1} x + \cos^{-1} y = \pi - \frac{2\pi}{3} = \frac{\pi}{3}$$

Answer: (B)

Q10. The value of $\tan^{-1}\left(\frac{1}{2}\right) + \tan^{-1}\left(\frac{1}{3}\right)$ is equal to:

(A) $\frac{\pi}{2}$ **(B)** $\frac{\pi}{4}$
(C) $\frac{\pi}{3}$ **(D)** $\tan^{-1}\left(\frac{5}{6}\right)$

Sol. Q10. We use the addition formula for $\tan^{-1} x + \tan^{-1} y$: Check condition: $xy = \left(\frac{1}{2}\right)\left(\frac{1}{3}\right) = \frac{1}{6} < 1$. Since $xy < 1$, the standard formula applies without adding π:

$$\tan^{-1}\left(\frac{1}{2}\right) + \tan^{-1}\left(\frac{1}{3}\right) = \tan^{-1}\left(\frac{\frac{1}{2} + \frac{1}{3}}{1 - \left(\frac{1}{2}\right)\left(\frac{1}{3}\right)}\right) = \tan^{-1}\left(\frac{\frac{5}{6}}{1 - \frac{1}{6}}\right) = \tan^{-1}\left(\frac{\frac{5}{6}}{\frac{5}{6}}\right) = \tan^{-1}(1) = \frac{\pi}{4}$$

Answer: (B)

Practice Questions — Part 3 (Q11 to Q15)

Q11. The numerical value of $\cos\left(2\tan^{-1}\left(\frac{1}{3}\right)\right)+\sin\left(2\tan^{-1}\left(\frac{1}{2}\right)\right)$ is equal to:

(A) $\frac{8}{5}$ **(B)** $\frac{7}{5}$
(C) 1 **(D)** $\frac{16}{25}$

Sol. Q11. We use the double-angle conversion formulas for $2\tan^{-1}x$:

1. $\cos\left(2\tan^{-1}\left(\frac{1}{3}\right)\right)=\frac{1-\left(\frac{1}{3}\right)^2}{1+\left(\frac{1}{3}\right)^2}=\frac{1-\frac{1}{9}}{1+\frac{1}{9}}=\frac{\frac{8}{9}}{\frac{10}{9}}=\frac{4}{5}$.
2. $\sin\left(2\tan^{-1}\left(\frac{1}{2}\right)\right)=\frac{2\left(\frac{1}{2}\right)}{1+\left(\frac{1}{2}\right)^2}=\frac{1}{1+\frac{1}{4}}=\frac{1}{\frac{5}{4}}=\frac{4}{5}$.

Sum $=\frac{4}{5}+\frac{4}{5}=\frac{8}{5}$. **Answer: (A)**

Q12. The simplest form of the expression $\tan^{-1}\left(\frac{\cos x-\sin x}{\cos x+\sin x}\right)$, where $-\frac{\pi}{4}<x<\frac{3\pi}{4}$, is:

(A) $\frac{\pi}{4}+x$ **(B)** $\frac{\pi}{4}-x$
(C) $x-\frac{\pi}{4}$ **(D)** $\frac{\pi}{2}-x$

Sol. Q12. Divide numerator and denominator inside by $\cos x$: $\frac{\cos x-\sin x}{\cos x+\sin x}=\frac{1-\tan x}{1+\tan x}=\tan\left(\frac{\pi}{4}-x\right)$. Given $-\frac{\pi}{4}<x<\frac{3\pi}{4}\Rightarrow-\frac{\pi}{2}<\frac{\pi}{4}-x<\frac{\pi}{2}$. Since $\frac{\pi}{4}-x$ lies safely in the principal branch $\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$, the inverse cancels directly: $=\frac{\pi}{4}-x$. **Answer: (B)**

Q13. On the set $\mathbb{Q}-\{-1\}$ of rational numbers, a binary operation $*$ is defined by $a*b=a+b+ab$. The identity element and the inverse of an element a are respectively:

(A) $e=0$ and $a^{-1}=\frac{-a}{1+a}$ **(B)** $e=1$ and $a^{-1}=\frac{a}{1-a}$
(C) $e=0$ and $a^{-1}=-a$ **(D)** $e=-1$ and $a^{-1}=\frac{1}{a}$

Sol. Q13.

1. **Identity:** $a*e=a\Rightarrow a+e+ae=a\Rightarrow e(1+a)=0\Rightarrow e=0$ (since $a\neq-1$). Check: $0*a=0+a+0=a$ ✓.
2. **Inverse:** $a*b=e=0\Rightarrow a+b+ab=0\Rightarrow b(1+a)=-a\Rightarrow b=\frac{-a}{1+a}$ (well-defined in $\mathbb{Q}-\{-1\}$).

Answer: (A)

Q14. The domain of definition of the real-valued function $f(x)=\sin^{-1}(2x-1)+\sqrt{1-x^2}$ is:

(A) $[-1,1]$ **(B)** $[0,1]$
(C) $\left[0,\frac{1}{2}\right]$ **(D)** $[-1,0]$

Sol. Q14. The domain is the intersection of its two components:

1. For $\sin^{-1}(2x-1)$: $-1\leq2x-1\leq1\Rightarrow0\leq2x\leq2\Rightarrow0\leq x\leq1$ ($D_1=[0,1]$).
2. For $\sqrt{1-x^2}$: $1-x^2\geq0\Rightarrow x^2\leq1\Rightarrow-1\leq x\leq1$ ($D_2=[-1,1]$).

Intersection: $D_1\cap D_2=[0,1]\cap[-1,1]=[0,1]$. **Answer: (B)**

Q15. If $\sin^{-1}x+\sin^{-1}y+\sin^{-1}z=\frac{3\pi}{2}$, then the value of $x^{100}+y^{100}+z^{100}-\frac{9}{x^{101}+y^{101}+z^{101}}$ is:

(A) 3 **(B)** 1
(C) 0 **(D)** -1

Sol. Q15. Since the maximum value of $\sin^{-1}t$ on $[-1,1]$ is $\frac{\pi}{2}$, the sum $\sin^{-1}x+\sin^{-1}y+\sin^{-1}z=\frac{3\pi}{2}$ forces each term to independently equal $\frac{\pi}{2}$: $\sin^{-1}x=\sin^{-1}y=\sin^{-1}z=\frac{\pi}{2}\Rightarrow x=y=z=1$. Substituting into the expression: $(1)^{100}+(1)^{100}+(1)^{100}-\frac{9}{(1)^{101}+(1)^{101}+(1)^{101}}=3-\frac{9}{3}=3-3=0$. **Answer: (C)**